Light Collection in a Polished Plastic Scintillator with Side-Mounted SiPMs

A Geant4 matrix study modelled on Rao *et al.*, J. Appl. Phys. **138**, 114504 (2025)

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G4DCSmonitor — Geant4 + ROOT simulation

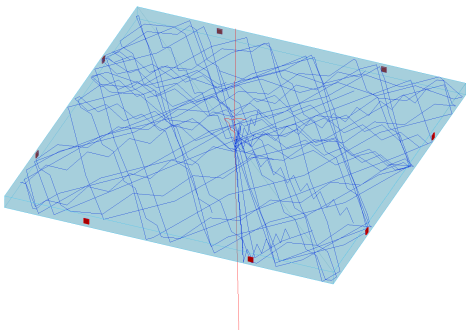
October 18, 2026

Outline

1. Aims: reproducing Rao *et al.*
2. Simulated detector
3. Optical inputs and SiPM model
4. Config 4: WLS fibre clamp
5. Results

Aims and reference measurement

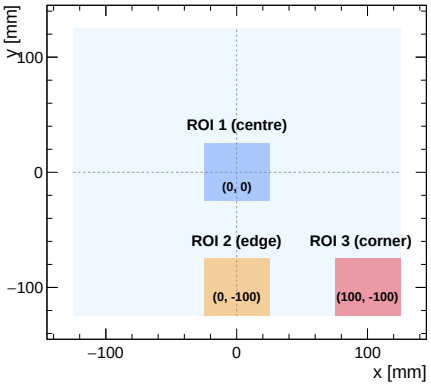
- Model the muon-trigger detector of Rao *et al.* (2025): polished 250 mm×250 mm×10 mm UPS-92S plate, side-coupled SiPMs.
- Question: how much of the *ordering* of light-collection efficiency over wrappers and layouts does optical transport explain?
- Deliverable: full matrix scan — 4 SiPM layouts × 4 reflectors × 3 beam spots, plus whole-plate, bare-contact and Al-backing systematics.
- Total **601 000 events**: 120k matrix + 400k whole-slab + 63k contact + 18k Al.



One vertical μ in glossy plate, cfg-1. Red = muon, blue = optical photons (display only).

What we compare against Rao

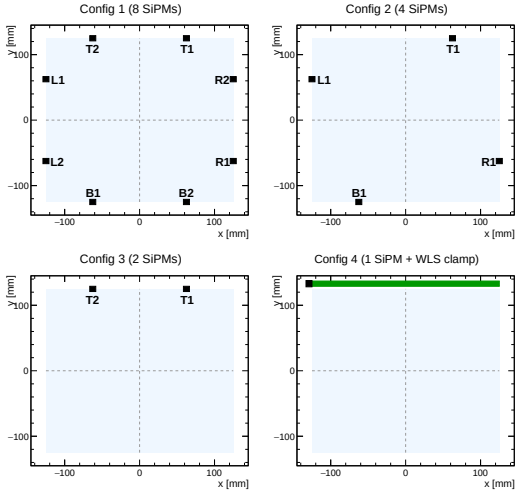
- Rao reports voltage pulse-height spectra (Fig. 10) for centre / edge / corner ROIs and 4 SiPM configurations.
- Our analogue: photoelectrons/event spectra in the same 3×4 grid (`fig_pe_grid`).
- Key experimental orderings to test:
  - **Reflectors:** bare $\ll$ TiO_2 $<$ Al $\approx$ glossy (glossy/Al best).
  - **Position:** corner lowest, centre highest (paper rule).
  - **Configs:** cfg-4 (WLS) competitive with direct readout.
- Result preview: we reproduce glossy brightest + TiO_2 weakest, but **not** bare-weakest nor corner-lowest everywhere.



Our three 50 mm×50 mm ROIs: centre (0,0), edge, corner — equal area, fully on plate.

Detector layouts (configs 1–4)

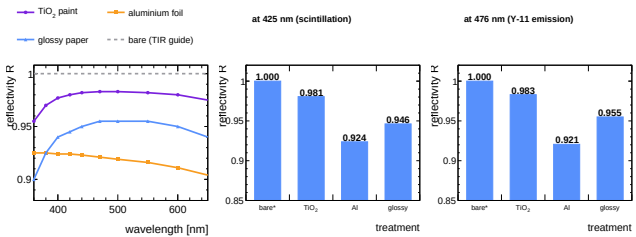
- Plate: PVT 250 mm×250 mm×10 mm, $n = 1.58$, polished, EJ-200/UPS-923A tables.
- SiPM: 4×4 mm epoxy box ($n = 1.55$) + 0.1 mm grease pad ($n = 1.46$).
- **Cfg-1:** 8 SiPMs, 2 per side at ±50% half-length.
- **Cfg-2:** 4 SiPMs, 1 per side at same offsets (C4-symmetric).
- **Cfg-3:** 2 SiPMs, both on $-x$ face (report top edge).
- **Cfg-4:** 1 SiPM + WLS fibre clamp on $-x$ face (next slide).



Schematics: black squares = SiPMs; green bar = cfg-4 clamp + readout SiPM at top-left.

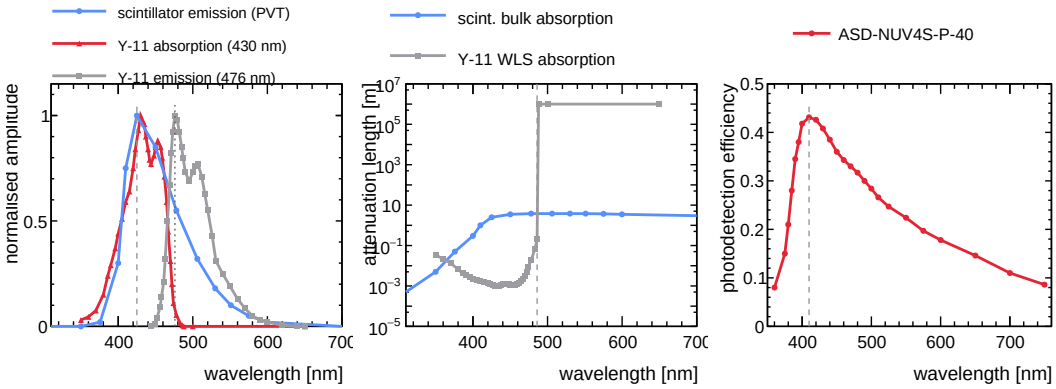
Reflector treatments: the heart of the model

	Bare	TiO_2	Al foil	Glossy
Contact	1.2 mm air gap	direct	grease / air	air gap
G4 finish	none (Fresnel/TIR)	groundfrontpainted	polishedbackpainted	polishedbackpainted
$R(\lambda)$	— (TIR guide)	0.955–0.983	0.904–0.925	0.900–0.955
Character	lossless guide	Lambertian, no TIR	specular, TIR + mirror	lossier mirror



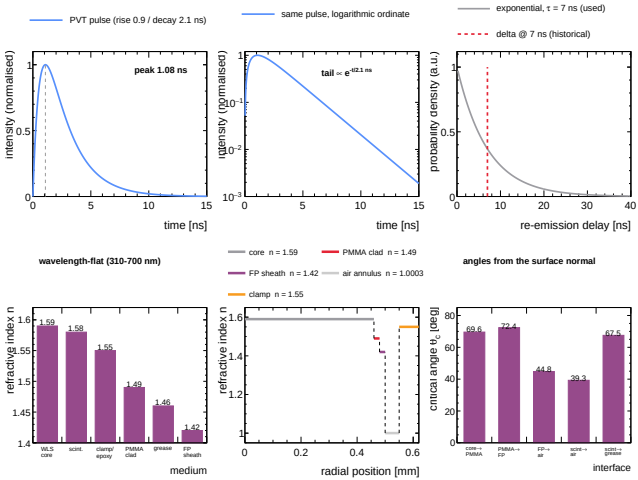
- Bare: no skin; sub-critical light killed in black holder ($R = T = 0$ skin $+ 1\,\mu$m bulk).
- Optional bare-contact fraction f ($R = 1 - f$): frustrated-TIR bracket.
- Al backing switch: `grease` ($n = 1.46$) vs `air` ($n = 1.0003$) — changes θ_c, not R.

Spectral inputs actually fed to Geant4



Left: scintillator emission (425 nm), Y-11 absorption (430 nm) + re-emission (476 nm). Middle: bulk vs WLS attenuation (WLS $L = 1.0$ mm at 430 nm). Right: ASD-NUV4S-P-40 PDE, peak 0.431 at 410 nm. All digitized tables, linearly interpolated.

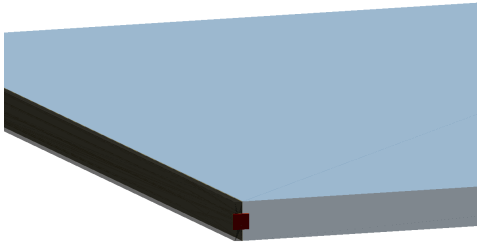
Timing, index ladder, and SiPM detection



- Scintillation: $f(t) \propto (1 - e^{-t/0.9})e^{-t/2.1}$ ns, peak at 1.08 nm; WLS re-emission exponential $\tau = 7$ nm (not fixed delay).
- Fibre staircase 1.59/1.49/1.42 + air annulus sustains core–cladding TIR.
- SiPM = `dielectric_metal` border with `EFFICIENCY` = PDE(λ), $R = 0$; `G4OpBoundaryProcess` draws PDE, sets `Detection`, invokes SD — no manual draw; SD counts 1 pe per hit.
- Per-event accounting: N_γ produced vs N_{pe} detected; `sipmHits` rows carry t_{first}, $\langle\lambda\rangle$, per-pe t/λ/creator vectors.

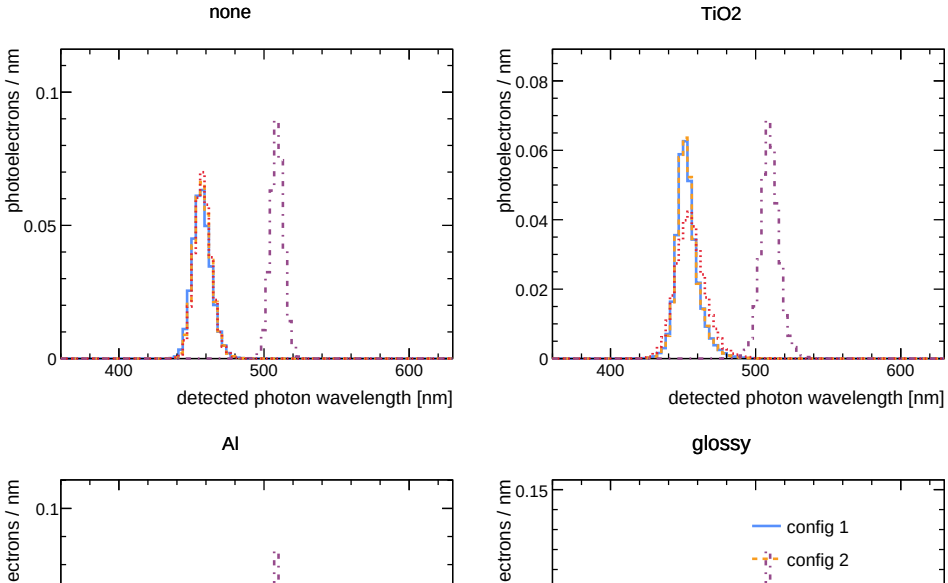
Config 4 close-up: analytic multi-clad Y-11 bundle

- 4 $\times$ Ø1.0 mm Kuraray Y-11(200) fibres, full 250 mm along $-x$ face.
- 2.5 mm pitch $\rightarrow$ 1.0 mm touching bundle over 41 mm convergence; analytic mitered `G4CutTubs` legs (no stuck tracks).
- Core $n = 1.59$ / PMMA 1.49 / sheath 1.42; 0.05 mm air annulus in Ø1.1 mm channels (deliberate!).
- 0.10 mm grease film; white Lambertian clamp ($R = 0.85$) on all faces; far-end Al mirror caps with explicit skin (fixes silent `NoRINDEX` kill, +40% centre).
- WLS: digitized Y-11 absorption + emission, $E'_\gamma \leq E_\gamma$; $\tau = 7$ nm exponential.

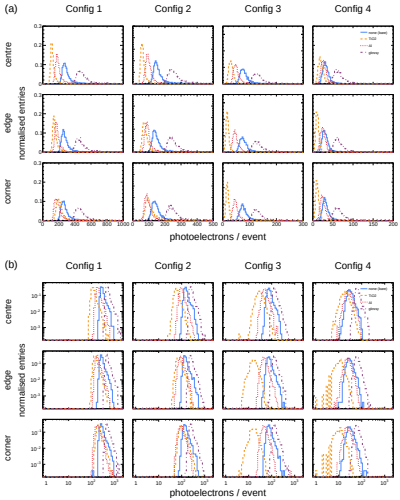




WLS works: detected spectrum green-shifts by 52 nm

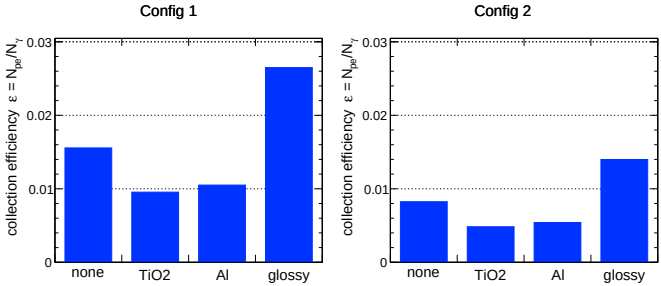


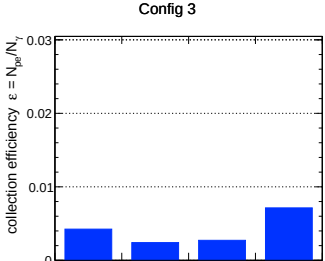
Photoelectron spectra: Rao Fig. 10 analogue

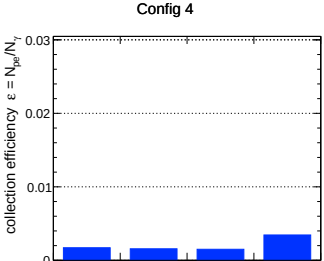


Rows = centre/edge/corner, columns = cfg 1–4, colours = wrappers. Ordering TiO_2 < Al < bare < glossy in every direct-coupled cell (cfg-1 centre: 125 / 200 / 304 / 521 pe). Cfg-4 an order of magnitude dimmer. TiO_2

Plate-averaged efficiency (whole-slab, 25k events/cell)



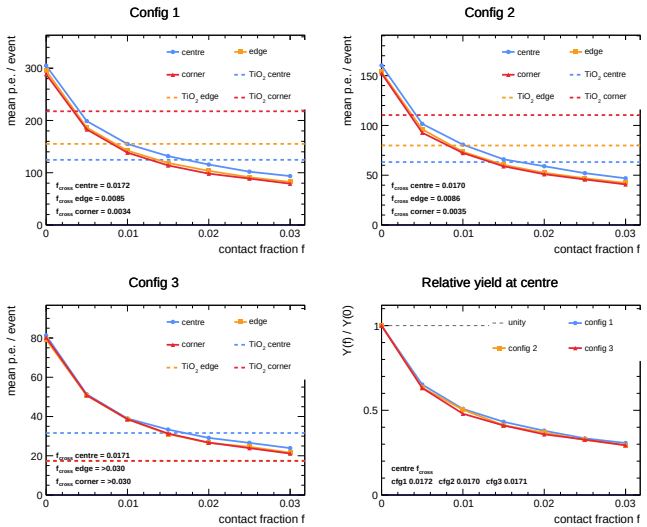




	Bare	TiO_2	Al	Glossy
Cfg-1	1.56	0.95	1.05	2.65
Cfg-2	0.82	0.48	0.54	1.40
Cfg-3	0.42	0.24	0.27	0.71
Cfg-4	0.17	0.16	0.15	0.35

$\varepsilon \times 10^{-2}$. Scales $\approx$ with sensor area; cfg-4 $\approx$1.5–2.5$\times$ below cfg-3. Cf. report Fig. 16 discussion.

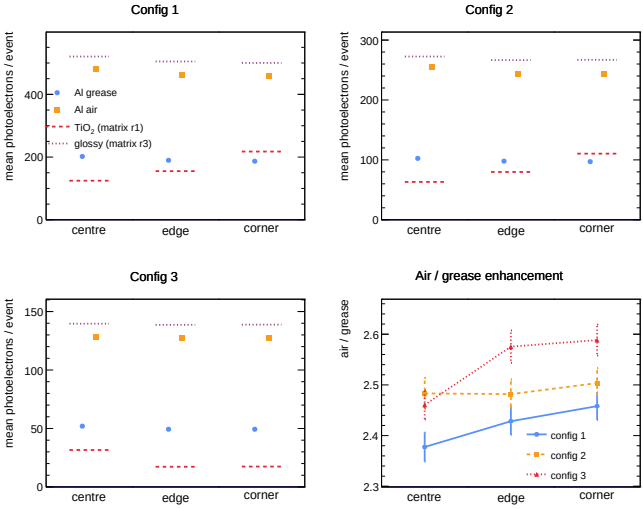
Why bare is bright: contact-fraction scan



- Clean gap ($f = 0$) = near-lossless TIR guide: cfg-1 centre 305 pe.
- $f = 0.01$ already halves light (155 pe); $f = 0.03 \rightarrow 94$ pe.
- Centre crossing $f \approx 0.017$ in all layouts; edge/corner 0.003–0.009 (cfg 1–2), > 0.03 (cfg-3 far side).
- **No single** f matches every layout/position — kept as systematic bracket, default stays $f = 0$.

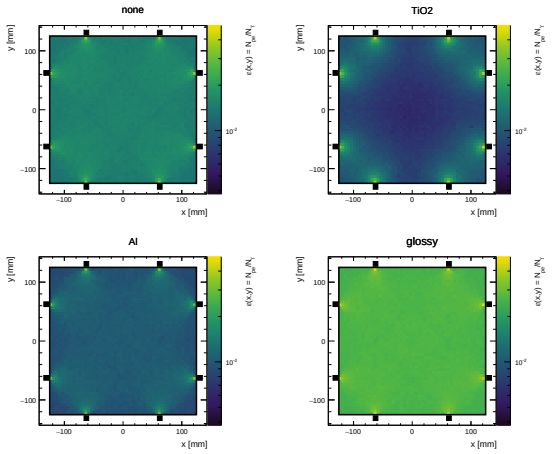
Cf. report Fig. 17.

Al backing: grease vs air brackets the real foil

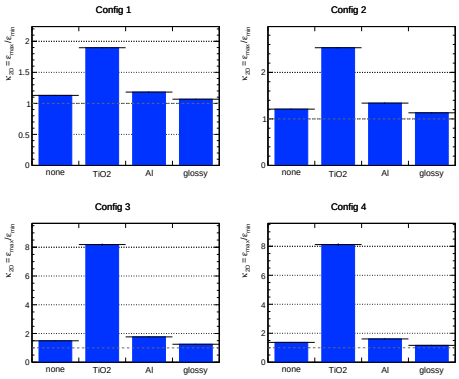


- Same $R(\lambda)$; only backing index changes θ_c: grease 67.5° vs air 39.3°.
- Air raises Al 2.38–2.59×: cfg-1 centre 202 → 480 pe.
- Air-backed Al ≈92 % of glossy — recovers Rao's Al ≈ glossy.
- Production whole-slab used grease (conservative lower branch).

Full-plate maps: glossy/bare flat, TiO_2 graded

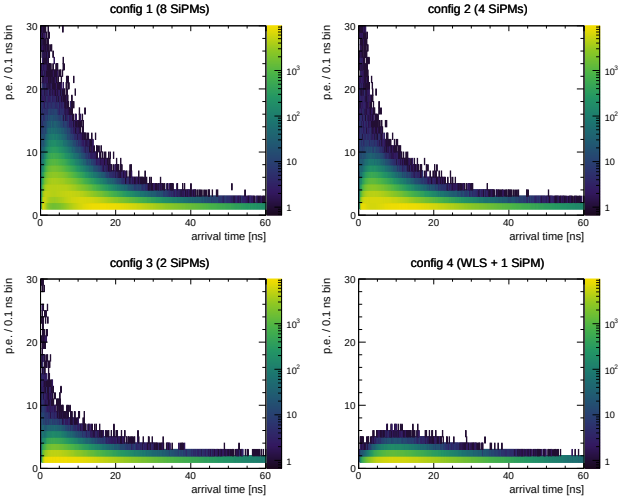


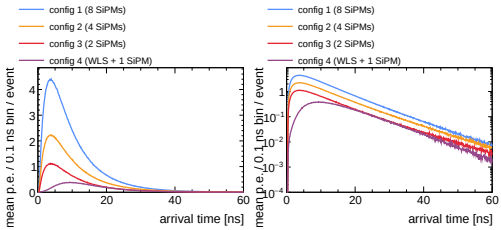
Cfg-1 maps, 5×5 mm cells, shared log-z. Bare/Al/glossy flat to $<9\%$; TiO_2 centre-dim (0.67% vs 1.02% outer, 1.51×).



Contrast $\kappa_{2D} = \max/\min$ over 50 mm cells. TiO_2 1.9–8.2, bare 1.13–1.49, Al 1.18–1.77, glossy 1.07–1.25. Cf. report Figs. 19, 21–23.

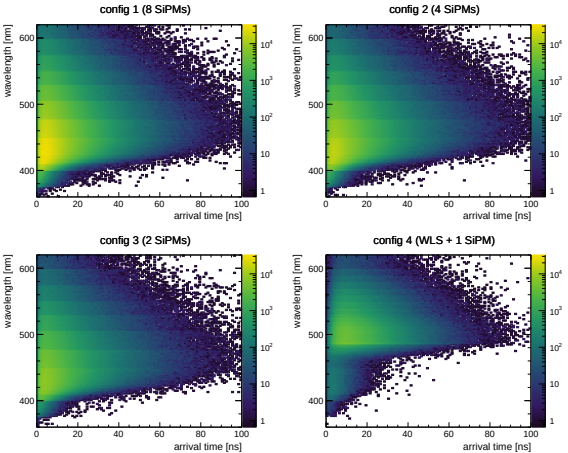
Pulses: persistence and average waveforms





Glossy, whole-plate. Modal arrival 3.8–4.0 ns (cfg 1–3) vs 9.25 ns (cfg-4); $\leq$5 nm fractions 32/31/30/7%; late (>20 nm) 11/12/13/25%. Integrals: 520/274/140/68 pe. Cf. Figs. 29–30.

Wavelength vs time + take-aways



Mean λ: 450.9 $\rightarrow$ 467.7 nm (cfg 1–3), 498.9 $\rightarrow$ 511.3 nm (cfg-4): late light reddens by self-absorption. Cf. Fig. 31.

What we reproduce

- Glossy brightest + most uniform in every layout; TiO_2 weakest (cfgs 1–3) and most position-dependent.

What we don't

- Bare is 2nd-brightest (TIR guide), not weakest; needs $f \approx 0.017$ contact to fall below TiO_2.
- TiO_2 corner-highest in cfg 1–2, vs paper corner-lowest.
- Cfg-4 $\approx$67 pe (glossy centre) $<$ cfg-3, vs paper cfg-4 $>$ cfg-3 (different normalisation/gain noted).

Bottom line: wrapper sets ordering, layout sets scale; f and Al-backing brackets quantify the systematics.